

What If Another Floyd?

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South Carolina

Abstract

The State of South Carolina is very vulnerable to the attack of hurricanes since it is adjacent with the Atlantic Ocean. In 1999 hurricane Floyd led most people living in the coastal region into a great panic. Take the city of Charleston as an example, 500,000 people rushed to Columbia, a safety heaven more than 200 kilometers away, through Interstate I-26. Since I-26 is the principal route going inland from Charleston to Columbia, there happened a standstill on the I-26. What is normally an easy two-hour drive took up to 18 hours to complete. Many cars simply ran out of gas along the way. Fortunately, Floyd turned north and spared the state this time, but the public outcry is forcing state officials to find ways to avoid a repeat of this traffic nightmare.

The principal proposal put forth to deal with this problem is the reversal of traffic on I-26, so that both sides, including the coastal-bound lanes, have traffic headed inland from Charleston to Columbia. We have set up a mathematical model to analyze this problem. Our conclusion is that the performance of the evacuation plan would be greatly improved. According to our estimation, it will take more than 40 hours for 500,000 people if the traffic on the coast bound lanes are not reversed. But after the lanes are reversed, only about 24 hours will be enough for 500,000 people, which is a great improvement. So we suggest that under very urgent situations all lanes of traffic should be forced to extend inland. With more exact calculations we suggest that if more highways of small scales extending inland are available, we not hesitate to open them and utilize all its inland bound lanes.

If people from more than one place wait for being evacuated, then another method should be presented. We may consider an alternative strategy, which staggers the evacuation, e.g. city by city, to further improve the evacuation plan's performance. Under certain conditions we notice that if all the people evacuate simultaneously it will lead to a disastrous result but that if they evacuate in turn they can avoid this. This phenomenon urged us to study what lies behind it. After extensive analysis we found out that it has something to do with the different time at which the hurricane will arrive in different places. A model is then set up and a recursive algorithm is proposed. This algorithm can help the decision maker to decide under what conditions a staggered strategy is called upon and what is the turn for evacuation of those cities.

If we go deep into this problem we should take the accommodating ability of those destination cities into consideration. For instance, the city of Columbia only has fewer hotel rooms than what is enough for the 500,000 or more people coming from the coastal area. People will simply choose to go on in their directions if they haven't got rooms. So the traffic outside Columbia will be greatly burdened. We suggest that the government of Columbia build more shelters, such as the temporary camps, to relieve the conflict. We can apply our model to work out the exact number of shelters that ought to be built given the total number of people to be evacuated.

We also pay attention to some details that are usually neglected by some people. First the type and the number of the vehicles on the road should be placed under strict restrictions to guarantee timely evacuation. Second we will need to make necessary changes to our plan if the evacuees from neighboring states come up to our state. The detailed changes depend on how earlier the neighboring states received the hurricane alarm than us.

1 Introduction

Evacuating the coast of South Carolina ahead of the predicted landfall of Hurricane Floyd in 1999 led to a monumental traffic jam. Traffic slowed to a standstill on Interstate I-26, which is the principal route going inland from Charleston to the relatively safe haven of Columbia in the center of the state. What is normally an easy two-hour drive took up to 18 hours to complete. Many cars simply ran out of gas along the way. Fortunately, Floyd turned north and spared the state this time, but the public outcry is pushing state officials to find ways to avoid a repeat of this traffic nightmare.

The principal proposal put forth to deal with this problem is the reversal of traffic on I-26, so that both sides, including the coastal-bound lanes, have traffic headed inland from Charleston to Columbia. Plans to carry this out have been prepared (and posted on the Web) by the South Carolina Emergency Preparedness Division. Traffic reversal on principal roads leading inland from Myrtle Beach and Hilton Head is also planned.

A simplified map of South Carolina is shown in the picture of the cover page. Charleston has approximately 500,000 people, Myrtle Beach has about 200,000 people, and another 250,000

people are spread out along the rest of the coastal strip. (More accurate data, if sought, are widely available.)

The interstates have two lanes of traffic in each direction except in the metropolitan areas where they have three. Columbia, another metro area of around 500,000 people, does not have sufficient hotel space to accommodate the evacuees (including some coming from farther north by other routes), so some traffic continues outbound on I-26 towards Spartanburg; on I-77 north to Charlotte; and on I-20 east to Atlanta. In 1999, traffic leaving Columbia going northwest was moving very slowly.

To solve this problem people aim at developing an effective model that would lead to practical strategies. Several important questions must be addressed. First, it is necessary for people to decide under what conditions should the lane be reversed. Then people have to find some measurements to evaluate the performance of different strategies. A staggered evacuation strategy should also be discussed in order to observe what extent it could improve the performance of the simple evacuation strategy. What's more, the model must be robust enough to accommodate many uncertain situations and sudden changes.

2 Assumptions and Symbols

Assumptions:

- Among all available highways, only those inland or coast bound are considered. Those ways parallel to the coast are not considered as evacuation routes.
- Since reversing traffic on a highway may bring additional cost (for instance, we need assign more police to administer the traffic), we will reverse the traffic on an additional lane only if the original lanes cannot carry their car flow.
- Normally the movement speed of a hurricane is quite slow. Thus, cars running on a highway at a speed bigger than the hurricane. Only those cars don't set off the coast will be affected by the landfall of the hurricane.

- Before the emergent evacuation is started and after the evacuation is over, the highways are considered to have no vehicle on them. Although they might actually carry a few vehicles on them, it is reasonable to omit them because their number is too small compared with the large car flow during the evacuation.
- During the emergent evacuation, the car flow on a highway can be viewed as a stable flow, in which all the vehicles are moving at a same speed and the distance between two adjacent vehicles are the same.
- Although various vehicles have various sizes, we assume that all vehicles have a same size, which can be determined by the average size of common vehicles.

Definitions:

- $P(t)$: the number of people that have been evacuated by the time t .
- N_0 : the number of people to be evacuated.
- N : the number of cars to be evacuated.
- Q : the flow of a highway, which means the number of vehicles passing through any section of this highway per hour.
- K : the vehicles density of a lane, which means the number of vehicles in a highway per kilometer in a stable condition. Its unit is km^{-1} .
- U : the speed of the cars on a highway.
- N_l : the number of lanes in one direction on a multi lane highway.
- T : the security time within which all the people must be evacuated. This security time is usually determined by the time interval between the prediction of the hurricane and its landfall.
- T_0 : the actual time it takes for all the N_0 people to evacuate from the coast.
- H : the average number of people per car during the emergent evacuation, $N = N_0/H$.

- L : the distance between two places.

3 Preliminary Analysis

Our task, known as the emergency evacuating problem, is to devise a strategy to evacuate all the coastal people (the total number is N_0) in a given period of time T . Let $P(t)$ be the number of people that can be evacuated within the given time T . Therefore if $P(T) \geq N_0$, the evacuation strategy is successful while it is unsuccessful if $P(T) \leq N_0$.

3.1 A Single Lane Model

First, we set up the single lane highway model. We usually use three important parameters to describe a single lane: the car flow Q , the car density K and the speed of the cars U . Their definitions have been given in the previous part. Considering their definitions, we know that $Q = KU$. According to Greenshields [4], we have

$$U = U_f(1 - \frac{K}{K_j}), \quad (1)$$

where U_f and K_j are parameters need to be determined. This linear formula Eq. (1) can describe the relationship between car density and speed quite well. We plot the function $U(K)$ in Fig. 1.

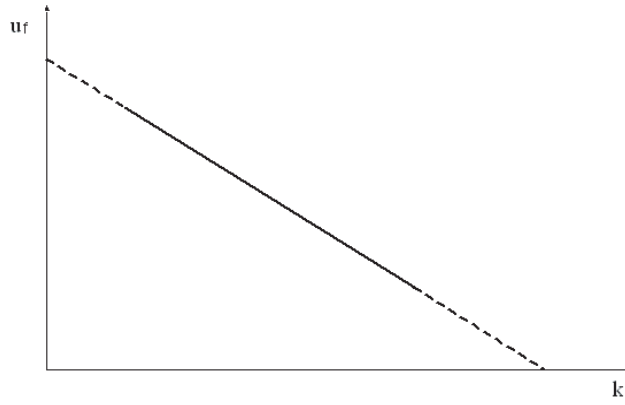


Figure 1: Function $U(K)$.

From practical data Greenberg proposed a more accurate model, in which he pointed out that when the car density exceeds a certain threshold, usually 75 km^{-1} , the relationship is logarithmic,

which is given by:

$$U = U_m \ln\left(\frac{K_j}{K}\right), \quad (2)$$

where U_m and K_j are parameters need to be determined.

When the car density is below 75 km^{-1} , Underwood pointed out that the relationship turns exponential, which is given by:

$$U = U_f \exp\left(-\frac{K}{K_m}\right), \quad (3)$$

where U_f and K_m are parameters need to be determined. We also plot the functions Eq. (2)(3) in Fig. 2.

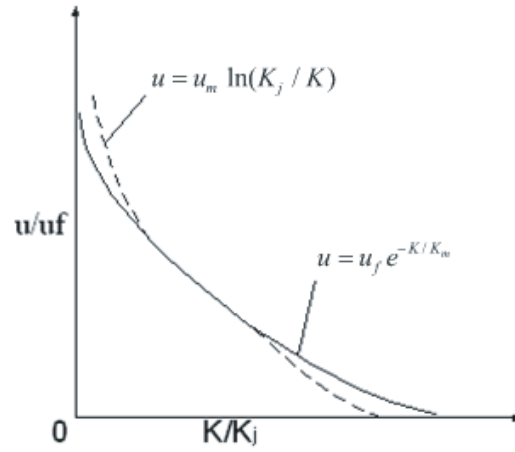


Figure 2: Functions $U(K)$.

We have found out the following data [1], which describe the relationship between a car's speed and the car head time gap defined as the time difference between two adjacent cars measured from head to head (see Tab. 1).

Table 1: Car's speed vs. car head time gap.

Speed (kph)	50	46	40	35	30
Time Gap (s)	2.13	2.16	2.20	2.26	2.33

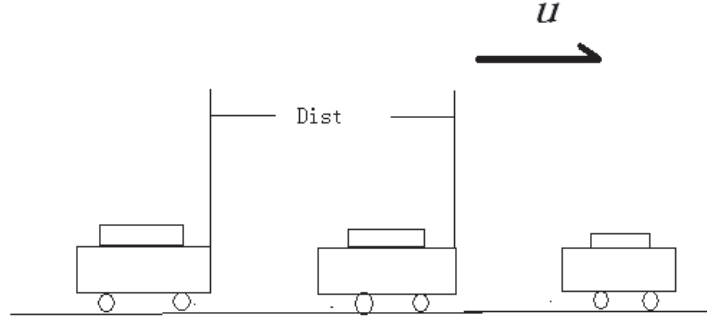


Figure 3: Distance between two adjacent cars.

Table 2: Car density.

Speed (kph)	50	46	40	35	30
Time Gap (s)	2.13	2.16	2.20	2.26	2.33
Dist (m)	30.0	27.6	24.5	22.0	19.5
Density (km^{-1})	33	36	41	45	51

We can use these data to calculate the distance between two adjacent cars (shown in Fig. 3), which is given by:

$$Dist = TimeGap * Speed. \quad (4)$$

Since we also have

$$Density = \frac{1}{Dist}, \quad (5)$$

we get out the car density under these five situations (see Tab. 2). We can see that all the five densities are below 75 km^{-1} , thus we are able to use the Underwood exponential model. We then use these five couples of data to determine the parameters U_f and K_m of the model: $U_f = 128.98 \text{ kph}$ and $K_m = 35.14 \text{ km}^{-1}$. The speed and the flow can be determined by:

$$U = 128.98 \exp\left(-\frac{K}{35.14}\right). \quad (6)$$

$$U = 128.98K \exp\left(-\frac{K}{35.14}\right). \quad (7)$$

Let $dQ/dK = 0$, then we can work out the maximum of Q , which is 1672 cars per hour (cph) while $K = K_m = 35.14 \text{ km}^{-1}$ and $U = U_m = 47.59 \text{ kph}$. It means at most 1672 cars can pass

through a single lane per hour. That is the largest road capacity of a lane. Thus we may calculate the total time it takes for all the coastal people to evacuate:

$$T_0 = \frac{N}{Q_{max}} = \frac{N_0}{Q_{max}H}. \quad (8)$$

We assume that a car takes 4 people ($H = 4$) on average and the Charleston has a population of 552,803 (N_0). Then we get $T_0 = 74.76$ hours.

3.2 A Multi-Lane Model

The above result is achieved only for a single lane highway. We learn from the problem that a common highway has two lanes in one direction. If we have N_l lanes in one direction we call this highway a $2N_l$ -lane highway. For instance an 8-lane highway has 4 lanes for one direction each. By extending our single lane model, we may easily derive a multi lane highway model. We then can calculate the total time now:

$$T_0 = \frac{N}{Q_{max}} = \frac{N_0}{Q_{max}HN_l}. \quad (9)$$

If we adopt the traffic reverse strategy and there are N_x inland bound lanes in all then T_0 is given by:

$$T_0 = \frac{N}{Q_{max}} = \frac{N_0}{Q_{max}HN_x}. \quad (10)$$

4 Development

4.1 Single-Source Single-Destination Evacuation Model

The simplest form of the evacuation problem is to evacuate people from one place to another place connected by a single highway. Applying our multi lane model, we may easily work out this problem. Take what happened in South Carolina during the 1999 Floyd Hurricane as an example. About 500,000 people in Charleston have to be evacuated to Columbia, which is about 225 *km* away, through the interstate highway I-26. Suppose a car can take 4 people ($H = 4$) and the city of Charleston has 552,803 ($N_0 = 552,803$) then we have got $T_0 = 37.38$ hours by Eq. (9) if no lanes are reversed.

Table 3: Reversed lanes vs. hours.

Reversed Lanes	T_0 (hour)
None Reversed	37.38
One Reversed	24.92
Two Reversed	18.69

Considering that I-26 has two coast-bounded lanes that can be reversed, T_0 can be further reduced by adopting the traffic reverse strategy (see Tab. 3). From Tab. 3 we learn that at least 18.69 hours are needed for all the people of Charleston to be completely evacuated.

We can modify this model for a more accurate result. Considering that the single lane model is based on an assumption that there exists a stable car flow on the road, we have to calculate the set-up time of the stable condition. We may suppose that at the beginning, the highway is empty thus the set up time T_{setup} will be:

$$T_{setup} = \frac{L}{U_{max}} = 4.73, \quad (11)$$

which is also the time an individual car will take in driving from Charleston to Columbia during the emergent evacuation. According to the 1999's report "What is normally an easy two-hour drive took up to 18 hours to complete", if we reverse traffic on two of the coast-bounded lanes of I-26, the situation will be greatly improved. More detailed plan of whether one lane or two lanes ought to be reversed depends on how long the security time T is.

We also notice at the end of the evacuation process the car flow on the highway cannot be viewed as a stable flow either. The highway with a stable flow will gradually turn empty at last. We do not have to include this period of empty time in the total time since we have already assumed in the previous part that every car already running on the highway will not be attacked by the hurricane because the hurricane moves more slowly than the car.

Thus the rectified total time T_r for all the people to be evacuated when no lanes are reversed will be:

$$T_r = T_0 + T_{setup} = 42.11. \quad (12)$$

4.2 Solution for Problem One

Therefore when the security time is longer than 42.11 hours it is not necessary to reverse traffic on the other 2 lanes. But when it is shorter than 42.11 hours, the traffic on the other two lanes should be reversed. Be aware that this result can only be achieved when the police strictly control the flow of a road at $1672 h^{-1}$ per lane.

5 Staggered Evacuation Strategy

In the previous model, only evacuation of the people in Charleston is considered. But we all know that in the reality almost all the cities along the coast are under the threat of hurricanes so we need to evacuate all of them. What's more, we should also take the hurricane's pattern into consideration. It means that the hurricane would arrive at different places at different time. Under such circumstances a staggered strategy may be more effective. We develop a recursive algorithm to determine under what conditions staggered strategy should be adopted and how the staggered strategy should be carried out.

As Fig. 4, suppose we are to evacuate people from n cities $C_1, C_2, \dots, C_n$ to a safety place D . Only city C_1 has a principle highway leading to city D , whose road capacity is Q (h^{-1}). The other $n - 1$ cities each has only one road to city C_1 . The population of each city is denoted by P_j . The security time of each city, denoted by T_i , varies since the hurricane will attack them at different times. Let $t(j)$ denote the time that city C_j needs to evacuate its entire people if all its people use the principle highway exclusively. It is given by $T(j) = P_j/Q$. Our task is to determine whether the staggered strategy should be adopted and, if it should be, in what sequence will the cities be evacuated.

Whatever strategy may be adopted, the total time for evacuating all the people remains unchanged:

$$T_{total} = \frac{\sum_{j=1}^n P_j}{Q}. \quad (13)$$

Our recursive algorithm goes through the following procedures:

Input. The cities to be sequenced

Output. Sequenced Cities

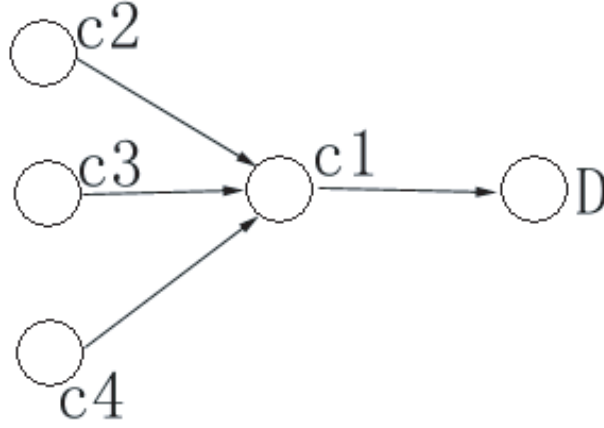


Figure 4: Staggered strategy.

Step 1. Check whether $T_j \geq T_{total}$ is satisfied for all the cities $\{C_j\}$. If satisfied, then it is certain that a staggered strategy is unnecessary. All strategies will be successful, go to Step 7.

Step 2. Check whether $t(j) \leq T_j$ is satisfied for all the cities $\{C_j\}$. If there exists a T_j that satisfies $t(j) > T_j$, then the people of city C_j will never be completely evacuated within its security time T_j . So any strategy will be unsuccessful, go to Step 7.

Step 3. Check how many cities satisfy $t(j) = T_j$. If there are two or more, one of them cannot be safely evacuated. So any strategy will be unsuccessful, go to Step 7. If there exists only one city satisfying the equation, then denote this city as city C_f and continue.

Step 4. Count the number of cities as A . Set priority value to be $1, 2, \dots, A$. The city with priority 1 will be evacuated first and the one with priority A will be evacuated at last. The priorities are assigned to the cities according to the following rules:

Rule 1. Assign priority 1 to city C_f if it exists.

Rule 2. All those cities satisfying $T_j \geq T_{total}$ will be assigned with priorities $A, A-1, A-2, \dots$.

The actual sequences of these cities themselves can be arbitrary.

If city C_f exists or at least one city satisfies $T_j \geq T_{total}$, then go to Step 5; otherwise any strategy will be unsuccessful, go to Step 7.

Step 5. If city C_f exists, let $T_j = T_j - T_f$ for all those unassigned cities C_j .

Step 6. Apply this algorithm again to those unassigned cities to get their sequences. Integrating

the output sequences of all the cities, we finally get the sequenced cities.

Step 7. End.

5.1 Solution for Problem Two

We may take the 1999 Hurricane Floyd as an example. The city of Charleston has a population of about 500,000, Myrtle Beach has a population of about 200,000. About 250,000 people are spread along the coast. We assume that those 250,000 live in two separate cities A and B with 125,000 in each one. Assume that the hurricane will arrive in city A in 10 hours. Charleston 25 hours, City B 30 hours, Myrtle Beach 36 hours. Only an interstate highway I-26 connects Columbia, a safe place and Charleston. Its road capacity is $1672 \times 4 = 6688$ (1/h) so the $t(i)$ for each city is that City A 4.67 hours, City B 4.67 hours, Charleston 18.7 hours and Myrtle Beach 7.48 hours. If we do not adopt the staggered strategy all the 950,000 will be evacuated in 35.5 hours, which means only Myrtle Beach can be evacuated safely. But if we adopt the staggered strategy with the sequence City A, Charleston, City B and Myrtle Beach, the four cities will be evacuated in 4.67 *h*, 23.4 *h*, 28.0 *h* and 35.5 *h*. Thus all of them can be evacuated safely adopting the staggered strategy (see Tab. 4).

6 Multi-Source Multi-Destination Evacuation Problem

6.1 Network Flow Model

Considering that we actually have more than one highway to use, we set up a net flow model catering to the reality. Under such circumstances people from several coastal places are to be

Table 4: Evacuation hours.

	City A	Charleston	City B	Myrtle Beach
Population	125,000	500,000	125,000	200,000
Security Time	10 <i>h</i>	25 <i>h</i>	30 <i>h</i>	36 <i>h</i>
<i>T</i>	4.67 <i>h</i>	18.7 <i>h</i>	4.67 <i>h</i>	7.48 <i>h</i>

evacuated in order to avoid the hurricane. Their destinations are inland places, which are able to hold them. This evacuation problem can be called multi-source-multi-destination evacuation problem.

We can use a network as Fig. 5 to denote the actual routes. Here a highway can be viewed as a combination of separate road pieces, which are divided by interchanges. A road piece is denoted by a single edge in the network. Each edge has a direction leading inland. Vertices in the network denote interchanges of highways. Besides, the source and destination cites are also denoted by vertices. Each edge has three parameters: *length* of this road piece, the *maximum car flow* which the highway can carry and the *actual car flow* the highway carries. We use P_i to denote the vertices and E_i to denote the edges. We are to determine the maximum number of people that can be evacuated through this network within time t . To do this we may work out the maximum flow in this network, which means the maximum of the flow that passes to the destination while it satisfies the restriction of each edge's capacity. We derive a new algorithm from the traditional Ford-Fulkerson algorithm to work out the maximum flow of the network. We have made a few modifications. For instance, we treat the forward edge and backward edge the same. Another modification we introduced is dynamic schema, which always promises us to get the maximal flow at a short time.

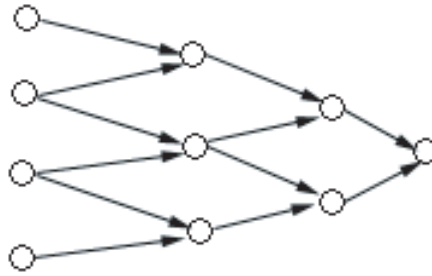


Figure 5: The network flow model.

We now define a new term “lane equivalent”, which we use “*l.e.*” in short. This term is a unit that describes the road flow on a multi-lane highway. 1 *l.e.* equals the maximum flow on a single lane, which is 1672 *cars/h*. Thus 1.6 *l.e.* means that the car flow is 1672*1.6=2675.2 *cars/h*. Notice that 1.6 *l.e.* does not mean that we will use 1.6 lanes on a multi lane highway (the number

of used lanes must be an integer) but that the flow is about 2675 cars/h . Virtually we will use 2 lanes to accommodate 1.6 *l.e.*, each lane having 1338 cars/h (0.8 *l.e.*).

If we adopt the traffic reverse strategy we may have up to 4 lanes on one way extending inland. Then the road capacity will be 4 *l.e.*. The actual number of lanes that will be reversed is determined by the result of the improved maximum flow algorithm. We use L_i to denote the actual flow that will be distributed on edge E_i . Then the traffic reverse strategy will be:

If $L_i \leq 2$ *i.e.* then we don't have to adopt the traffic reverse strategy. The original lanes can be used.

If $2 \text{ l.e.} < L_i \leq 3 \text{ l.e.}$ then we should adopt the traffic reverse strategy. One lane will be reversed.

If $3 \text{ l.e.} < L_i \leq 4 \text{ l.e.}$ then we also should adopt the traffic reverse strategy. Two lanes will be reversed.

Let us take the actual map of SC as an example. The result is shown in Fig. 6.

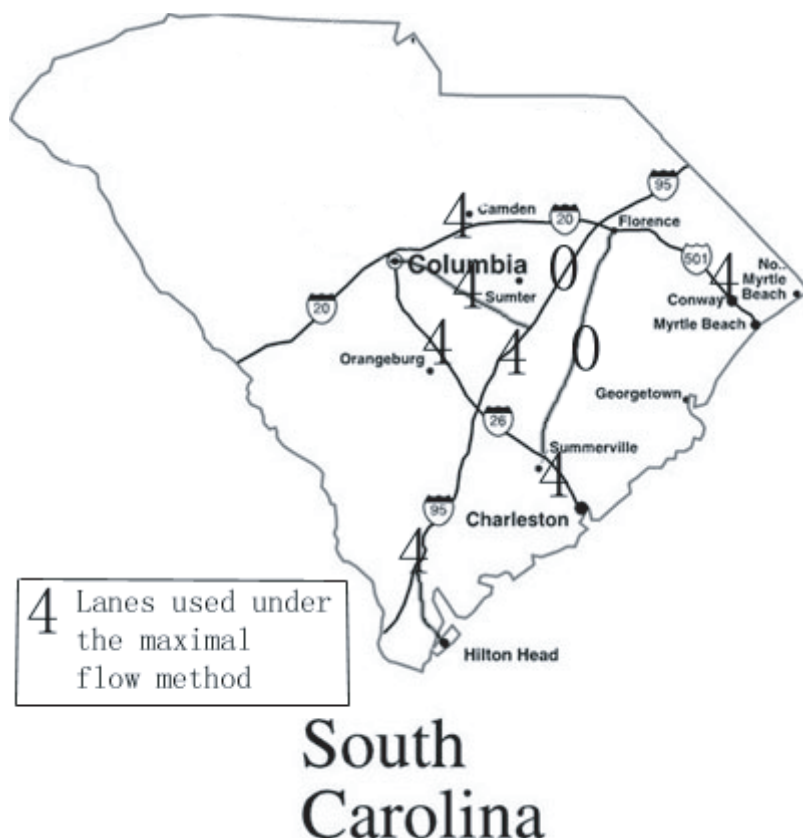


Figure 6: The maximal flow method.

6.2 Solution for Problem Three

In the actual situation, several smaller highways besides I-26 extend in land from the coast. If we turn some traffic around onto these roads the performance of the evacuation may be improved. We can use our network flow model to describe the situation. First we assume that these highways are not available. We calculate the maximal car flow the original network can carry. And then we add these smaller highways into our network and calculate the maximal network flow again. If the result is larger than the previous one, it is worthwhile to open these highways. If not then it is a waste to open them. That is how we decide whether the extra highways should be turned to use or not.

6.3 Solution for Problem Four

In the previous discussion we neglect the actual accommodating ability of the destination city. We had assumed that the destination city can hold as many people as there are, but it is not true in the reality. For instance, Columbia does not have sufficient hotel space to accommodate the evacuees (including some coming from farther north by other routes), so some traffic continues outbound on I-26 towards Spartanburg; on I-77 north to Charlotte; and on I-20 east to Atlanta. In 1999, traffic leaving Columbia going northwest was moving only very slowly. One way to reconcile the conflict is to build more shelters in Columbia. Our network flow model can also be used to determine to what extent it can reduce the traffic leaving Columbia.

First, we introduce a new term “virtual route”. In our previous model the vertices that stand for interchanges can accommodate nothing so the inward flow must equal the outward flow at such vertex. For Columbia since it is the destination vertex, it only has inward flow but no outward flow. In this question Columbia will be not be considered as a destination vertex but also an interchange. The network flow model allows no accommodation for a vertex so we must use a “virtual route” to “transfer” all the people in the shelters. This virtual route extends from Columbia to another “virtual city”. Its maximal road capacity equals the number of people accumulating in the shelters per hour (see Fig. 7).

With the concept of “virtual route” we may solve Requirement Four. We first add this virtual

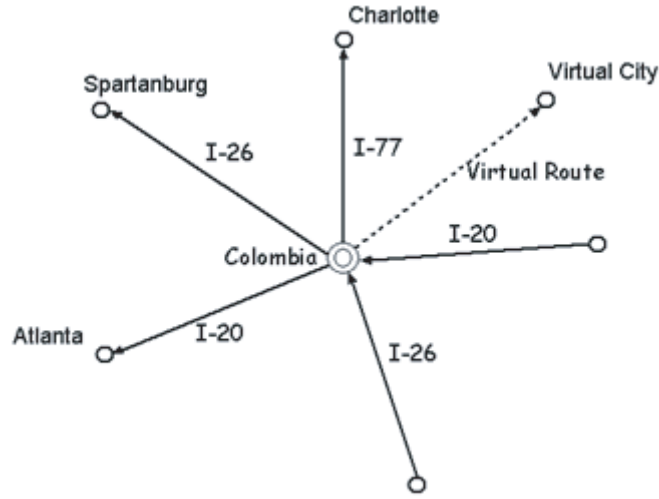


Figure 7: Virtual route.

route and the virtual city into our network and then calculate the maximal flow the network can carry again. We can also know the car flow on the virtual route and the total time it takes for all the people to evacuate. Multiply the two terms and then we get the accommodating ability of the shelters. The actual shelters built can not be smaller than this number.

6.4 Solution for Problem Five

In 1999, many families leaving the coast brought along their boats, campers, and motor homes. Many drove all of their cars. So under certain conditions there should be restrictions on vehicle types or numbers of vehicles brought in order to guarantee timely evacuation.

Using our network flow model and the Ford-Fulkerson algorithm we can work out the maximum flow of the network Q_{max} , which equals the number of cars arriving at the destination per hour. Suppose the security time is T_0 , the maximum cars that can be evacuated within T_0 are QT_0 . If $QT_0 < N_0/H$ then some people will not be evacuated timely. One way to solve this problem is to increase H , which means that a car should carry more people. For instance if the security time is 24 hours and the maximum flow of the network is 5000 ($1/h$) then the number of cars can be evacuated is $24 \times 5000 = 120,000$. Therefore if 500,000 people ought to be evacuated it must be ordered that a car must carry at least $INT[500,000/120,000] + 1 = 5$ people.

Another important factor that we should take into account is the type of vehicles. In our single

lane model when we work out the maximum of the car flow we also learn that the vehicles must move at 47.59 kph under such condition. But some vehicles cannot move so fast. For instance, motor homes or cars carrying very large items like boats or campers may not move at such a speed. Because of this, the government should enact a law that forbids those vehicles moving more slowly than 47.59 kph to evacuate. Considering that it is very difficult for a vehicle to maintain its peak speed at any time, a relaxation is allowed. So the speed lower bound can be 50 km/h .

6.5 Solution for Problem Six

Our model can be applied to solve this problem with a few changes. Considering that the Georgian car flow came up I-95 fleeing earlier predicted landfalls of Hurricane Floyd to the south, we may suppose that when the Georgian cars embarked on the highway, I-95, I-26 and other highways are all nearly empty, we may apply the improved maximal flow algorithm now to see how much road capacity is occupied by Georgian cars. And then subtract the occupied flow from those highways' maximal road capacity and apply the maximal flow algorithm again. The result can be viewed as the actual car flow when Georgian cars come. Compare this actual flow with the normal car flow when no Georgian cars come we can see how big the impact is.

7 Evaluate Different Evacuation Plans

The most important standard to evaluate an evacuation plan is whether it can evacuate all the people within the given security time. Of all our models we always stick to this standard, which means that we will sacrifice anything else to evacuate all the people before the landfall of the hurricane.

The other two reasonable standards are how many people a plan can evacuate per hour (the more the better) and how many lanes a plan will reverse (the fewer the better). It is hard to say which of the other standards is more important. They are opposite to each other. If we want to fit better to one standard the other must be sacrificed.

8 Strengths and Weaknesses

Although our tasks are to solve evacuation problems for the state of South Carolina we can see that our network flow model is very vigorous. It can deal with evacuation problems anywhere without any specific restriction. For instance if the highways in the network each has a different number of lanes we can adjust our model by changing the weight of each edge. We have developed a program that can devise the evacuation plan automatically after the principle routes, population of each city and the hurricane pattern are inputted into the computer. Thus we dare say that our model can be put into practical use.

We want to improve our model by a switching method. Actually the evacuation plan we get might not be the best. The evacuation time might not be the least in the multi-source multi-destination model. To get better results a more accurate piecewise model can be set up. In this model the maximal flow algorithm is reapplied every time a destination has been fully evacuated and the flow may be switched after this procedure. So making a few adjustments during the switching process is necessary. This modified algorithm can greatly improve the performance.

Our model performs quite well under various accidental situations. For instance it may deal with the sudden accident on a road. If a car has been smashed, all the cars behind the smashed car will be blocked. When it happens we only have to make a few changes to our network flow model, that is, to split the edge into two edges at the place where the smashed car is.

The weakness of our model is that it is not optimal for the specific SC map so it might not yield the best result for this certain problem. But it is the price of its generality.

9 Summary

In order to solve the evacuation problem, we first set up a single lane model to describe the traffic on a highway. A preliminary model to calculate the car flow on a single lane road is then developed. Further we extend this preliminary model to common multi-lane highways. This model can be used to work out the single-source single-destination evacuation problem.

To cater to the actual needs we build a more general multi-source multi-destination evacuation model. The core of this model is to propose a modified network flow algorithm appealing to the

traditional Ford-Fulkerson algorithm. We apply the proposed algorithm to the real Hurricane Floyd of 1999, achieving sufficiently satisfactory results. To make the model more realistic and to widen its range of applications, many aspects are considered. For instance a “virtual route” technique is introduced to figure out the shelter-building problem.

What’s more, a staggered evacuation method is discussed. In order to decide whether a staggered method should be adopted and how the staggered method is carried out, we develop a recursive algorithm to assign priorities to those cities. According to their priority the cities are evacuated one by one.

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